Sukrut Shishupal

PhD student, Department of Biomedical Informatics
University of Utah

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BIOGRAPHY

Sukrut Shishupal is a **PhD student** in the Department of Biomedical Informatics at the **University of Utah**, with a strong background in computer science, biomedical data science, and environmental health. His research focuses on temporal pattern mining and the development of shapelet-based methods to model high-resolution exposure data from air quality sensors and link these motifs to health outcomes.

He and his lab have published one of the first open-source, long-term EPA shapelet library spanning from 2004–2024 (https://ehie-shapelets.ctsi.utah.edu/), with scalable tools for motif discovery, imputation, and spatiotemporal analysis. His work integrates time-series mining, machine learning, and public health applications, with use cases in areas such as sports performance, stillbirth risk, and nursing task shifts. In addition to his methodological contributions, he actively collaborates across multidisciplinary teams, contributes to open-source codebases, and is committed to developing tools for environmental epidemiology.

Research interest

Interpretable AI, Shapelet-based time series analysis, environmental exposure modeling, spatiotemporal analytics, public health informatics, scientific computing

Professional Experience

University of Utah
Graduate Teaching Assistant — Since 08/2023
Graduate Research Assistant — Since 11/2022

Automation Teknix — 11/2020 – 05/2021
Trainee Software Developer

Centre for Materials and Electronics Technology — 06/2017 – 08/2017
Internship

Education

Doctoral Program in Biomedical Informatics — Since 01/2025
University of Utah
Advisors: Prof. Kathy Sward, Dr. Ramkiran Gouripeddi

Master's Program — 08/2022 – 05/2024
Department of Biomedical Informatics
Data science Track
Graduated with highest distinction

Integrated Bachelor's and Master's 08/2014 – 05/2020
Institute of Bioinformatics and Biotechnology
Savitribai Phule Pune University

PUBLICATIONS

1. Cummins M, **Shishupal S**, Wong B, Wan N, Han J, Johnny J, Mhatre-Owens A, Gouripeddi R, Ivanova J, Ong T, Soni H, Barrera J, Wilczewski H, Welch B, Bunnell B; Travel Distance Between Participants in US Telemedicine Sessions With Estimates of Emissions Savings: Observational Study; J Med Internet Res 2024;26:e53437; DOI: 10.2196/53437

2. Mollie R Cummins, Bob Wong, Neng Wan, Jiuying Han, **Sukrut D Shishupal**, Ramkiran Gouripeddi, Julia Ivanova, Asiyah Franklin, Jace Johnny, Triton Ong, Brandon M Welch, Brian E Bunnell, Social vulnerability, lower broadband internet access, and rurality associated with lower telemedicine use in U.S. Counties, JAMIA Open, Volume 8, Issue 4, August 2025, ooaf056, https://doi.org/10.1093/jamiaopen/ooaf056

Thesis Paper

Master's Thesis: Binding Affinity Prediction of Protein-Protein Complexes Using Machine Learning
Mentor: Dr. Sukanta Mondal
Publication date: May 2019

Bachelor's Thesis: Synthesis and Characterization of Molecularly Imprinted Polymers for Solid-Phase Extraction of Biomolecules
Mentor: Dr. Smita Zinjarde
Publication date: May 2017

Paper/Poster Talks

For slides, see https://sukrut-shishupal.github.io/talks/

1. Unlocking Patterns in Exposure Data with Shapelets
   Lightening talk, DELPHI symposium 2025

2. Exploring Patterns of Interstate Telemedicine Using Visualization Framework
   DELPHI symposium 2024

3. Data Science and Teamwork for Understanding PASC: PCORnet™ RECOVER EHR
   Data Science Day 2023

TEACHING

BMI 6018 – Introduction to Programming, University of Utah, Fall 2023
Teaching Assistant. Instructor: Dr. Ramkiran Gouripeddi
Responsibilities: co-developed teaching material, taught multiple lectures

BMI 6016 – Biomedical Data Wrangling, University of Utah, Spring 2024, Spring 2026
Teaching Assistant, Instructor: Dr. Ramkiran Gouripeddi
Responsibilities: co-developed teaching material, taught multiple lectures